INORGANIC CUMULATIVE EXAM

April 3, 1994

Answer either Part A <u>or</u> Part B.
A high level of performance is required to pass this cume.

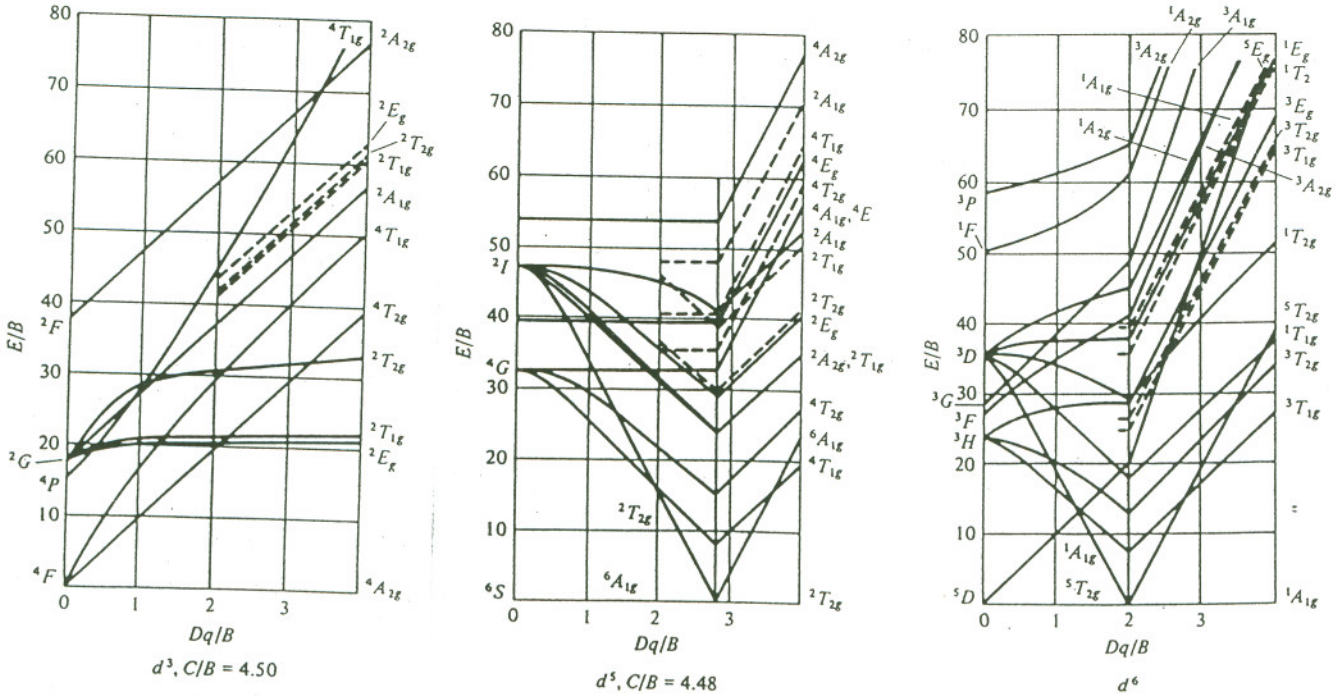
PART A

1) The complex $[Ni(CN)_4]^{-2}$ is diamagnetic but $[NiCl_4]^{-2}$ is paramagnetic with two unpaired electrons. Explain these experimental observations using simple ligand field or MO theory.

2) Both nickel and platinum belong to the same family on the periodic table, but the complexes $[NiCl_4]^{-2}$ and $[PtCl_4]^{-2}$ differ considerably in geometry, color and magnetism. Using any theory, discuss the magnetic, geometric and spectroscopic properties of these complexes.

3) Diamagnetic complexes of cobalt such as $[Co(NH_3)_6]^{+3}$, $[Co(en)_3]^{+3}$ and $[Co(NO_2)_6]^{-3}$ are orange-yellow. In contrast, the paramagnetic complexes $[CoF_6]^{-3}$ and $[Co(H_2O)_3F_3]$ are blue. Explain qualitatively this difference in color.

4) What is the ground state term symbol for Co^{+2}?



INORGANIC CUME

Using the attached figures and tables, answer eight of the following nine questions.

1. Does this complex obey the "18 electron rule"? Justify your answer.

2. Account for the differences in the Mo-C bond distances.

3. Account for the relationship between the M-C and C-O distances. Include in your comments an identification of the orbitals involved.

4. How might you explain the P-O-P bond angles?

5. Give logical values for the numbers which have been blacked out of table 1.

6. Describe what the P NMR spectrum might look like for this complex.

7. For the carbon NMR, how many different C atoms are in this complex?

8. Estimate the values for the blacked out values in the torsion angle table.

9. Explain the differences in the P(2)-O(6) and P(3)-O(6) distances.

Formula	$C_{34}H_{30}O_8P_3Mo$
MW, daltons	750.45
Space group	$P\bar{1}$
a(Å)	8.868(2)
b(Å)	11.552(2)
c(Å)	16.818(3)
α(°)	78.32(1)
β(°)	93.70(2)
γ(°)	86.59(2)
V(Å^3)	1679.0
Z (molecules/cell)	[redacted]
d_{calc} (g/cm^3)	1.512
Crystal dimensions, mm	[redacted] × [redacted] × [redacted]
h_{max}, h_{min}	15, 0
k_{max}, k_{min}	17, [redacted]
l_{max}, l_{min}	25, [redacted]
Refl. meas.	7233
Scan width (°)	1.2
Refl with $I \geq 3\sigma I$	6267
No. of variables	[redacted]
R, %	[redacted]

Table 5. Selected bond distances and angles and torsion angles for *cis*-$Mo(CO)_4\{(Ph_2PO)_2P(O)C_6H_4$-*p*-OMe$\}$, II

Atoms	Distance	Atoms	Distance	Atoms	Distance
Mo—P1	2.459(1)	P1—C6	1.825(5)	P3—O7	1.458(3)
Mo—P2	2.457(1)	P1—C7	1.810(4)	P3—C5	1.756(4)
Mo—C1	2.005(5)	P2—O6	1.664(3)	O1—C1	1.143(6)
Mo—C2	1.988(5)	P2—C8	1.818(4)	O2—C2	1.158(7)
Mo—C3	2.050(5)	P2—C9	1.819(5)	O3—C3	1.132(6)
Mo—C4	2.019(4)	P3—O5	1.591(3)	O4—C4	1.153(6)
P1—O5	1.660(3)	P3—O6	1.586(3)		

Atoms	Angle	Atoms	Angle	Atoms	Angle
P1—Mo—P2	91.33(4)	P2—Mo—C3	92.2(1)	Mo—P2—C9	121.8(1)
P1—Mo—C1	175.8(1)	P2—Mo—C4	87.0(1)	O5—P3—O6	104.9(1)
P1—Mo—C2	90.4(2)	Mo—P1—O5	114.9(1)	O5—P3—O7	114.5(2)
P1—Mo—C3	87.0(1)	Mo—P1—C6	122.4(2)	O5—P3—C5	102.1(2)
P1—Mo—C4	93.5(1)	Mo—P1—C7	115.8(1)	P1—O5—P3	128.8(2)
P2—Mo—C1	88.1(1)	Mo—P2—O6	116.7(1)	P2—O6—P3	130.2(2)
P2—Mo—C2	176.7(2)	Mo—P2—C8	115.2(1)		

Atoms	Torsion angle	Atoms	Torsion angle
P2—Mo—P1—O5	[redacted]	O7—P3—O5—P1	−56.4(3)
P1—Mo—P2—O6	16.1(1)	C5—P3—O5—P1	177.3(2)
Mo—P1—O5—P3	−34.4(3)	O5—P3—O6—P2	−58.1(3)
Mo—P2—O6—P3	17.0(3)	O7—P3—O6—P2	68.1(3)
O6—P3—O5—P1	[redacted]	C5—P3—O6—P2	−164.8(2)

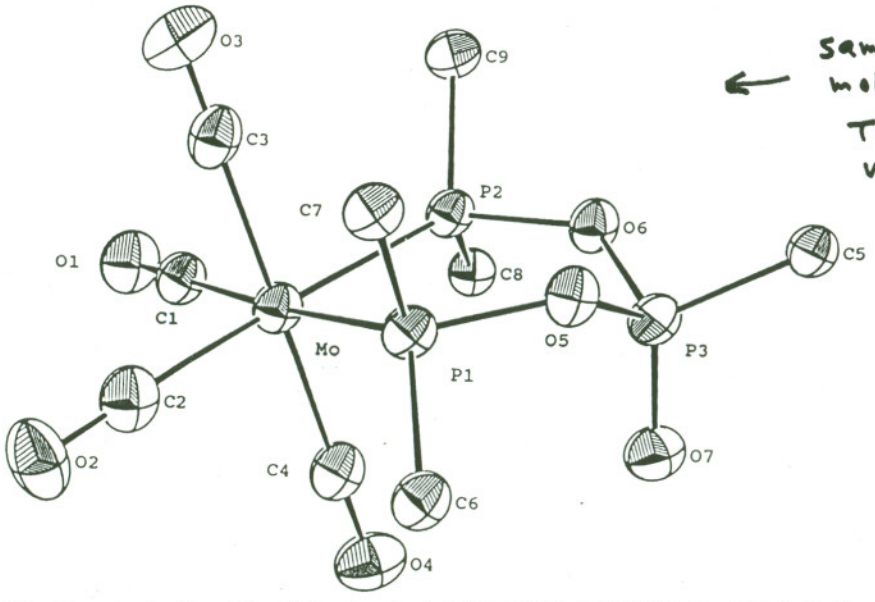


Fig. 4. ORTEP drawing of the chelate ring in *cis*-$Mo(CO)_4\{(Ph_2PO)_2P(O)C_6H_4$-*p*-OMe$\}$, II. The thermal ellipsoids are drawn at the 25% probability level and the hydrogens are omitted for clarity.

← same molecule, two views →

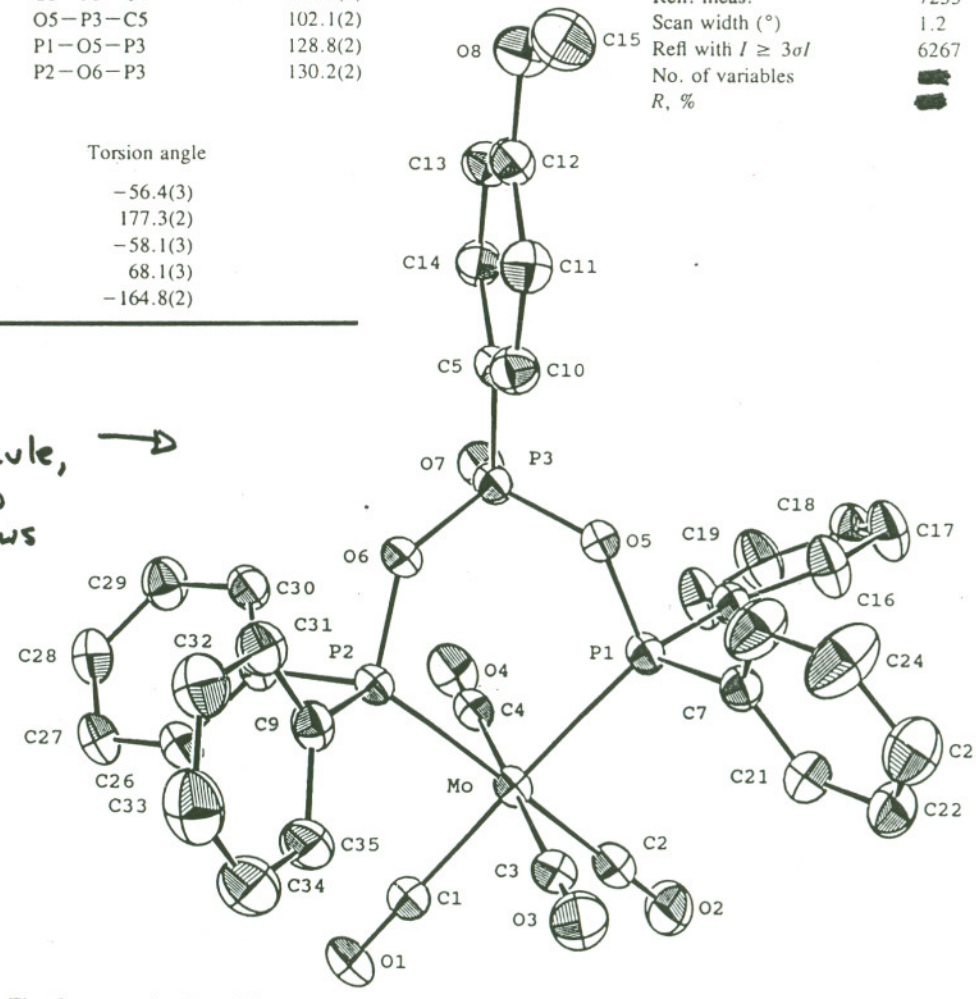


Fig. 2. ORTEP drawing of the molecular structure of *cis*-$Mo(CO)_4$-$\{(Ph_2PO)_2P(O)C_6H_4$-*p*-OMe$\}$, II. The thermal ellipsoids are drawn at the 25% probability level and the hydrogens are omitted for clarity.